



# Decarbonized Aviation: A Gendered Focused Examination

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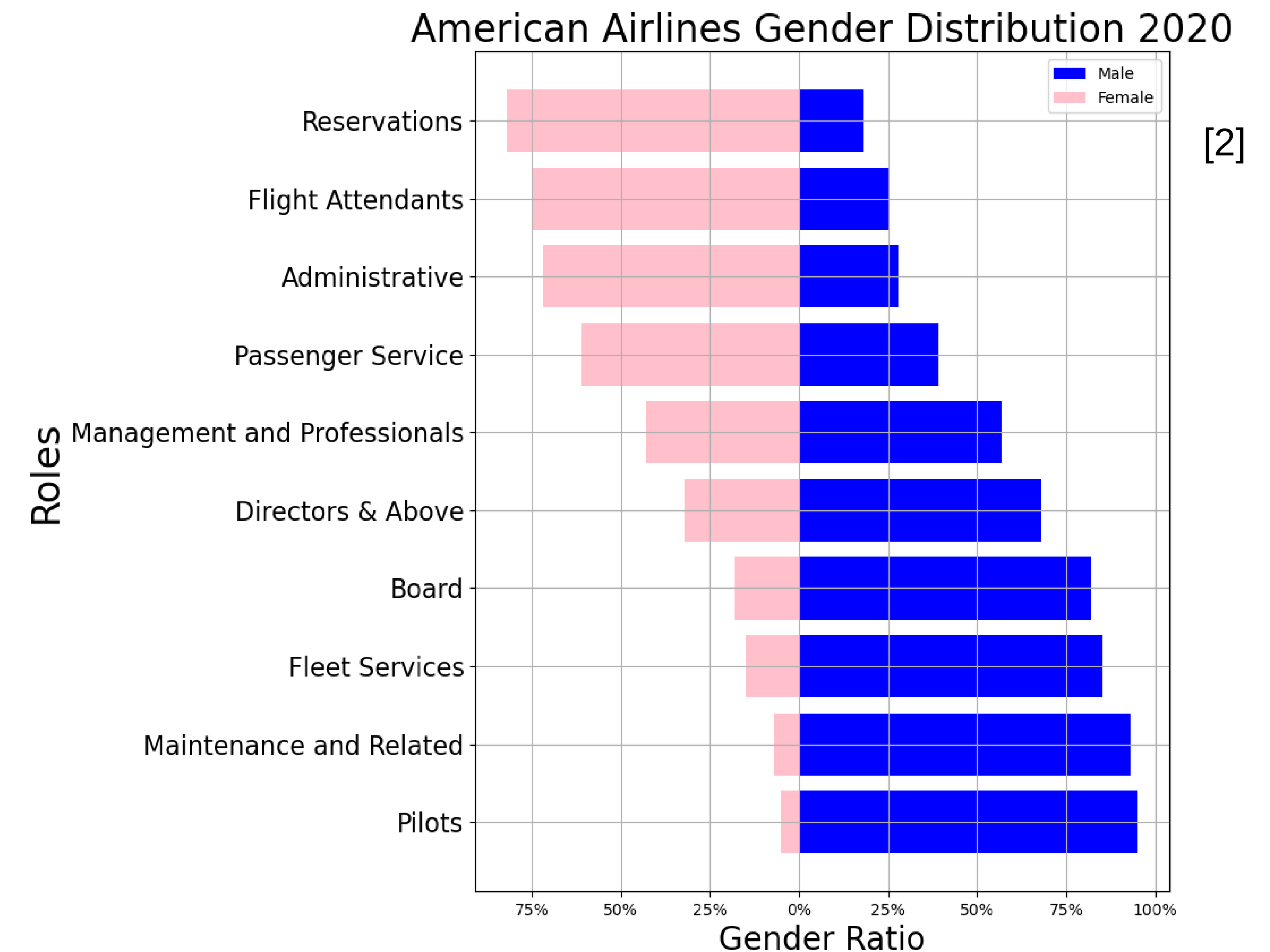
## Gendered Aviation

### Gender Disparity

- Women's participation in the labour force has significantly risen in the last few decades, yet they still remain underrepresented in high-wage and managerial positions across many industries.
- The underrepresentation of women in senior and technical roles within the aviation industry is a persistent issue which reflects boarder industry trends.
- In the last two decades, the ratio of women-to-male pilots has been relatively stagnant, only increasing by ~3%, despite attempts to encourage gender diversity [1].
- Facing the ongoing labor crisis, the aviation industry has a timely opportunity to address gender disparity and work towards diversity and inclusion by increasing women's participation, especially in senior and technical positions.

### Challenges and Barriers

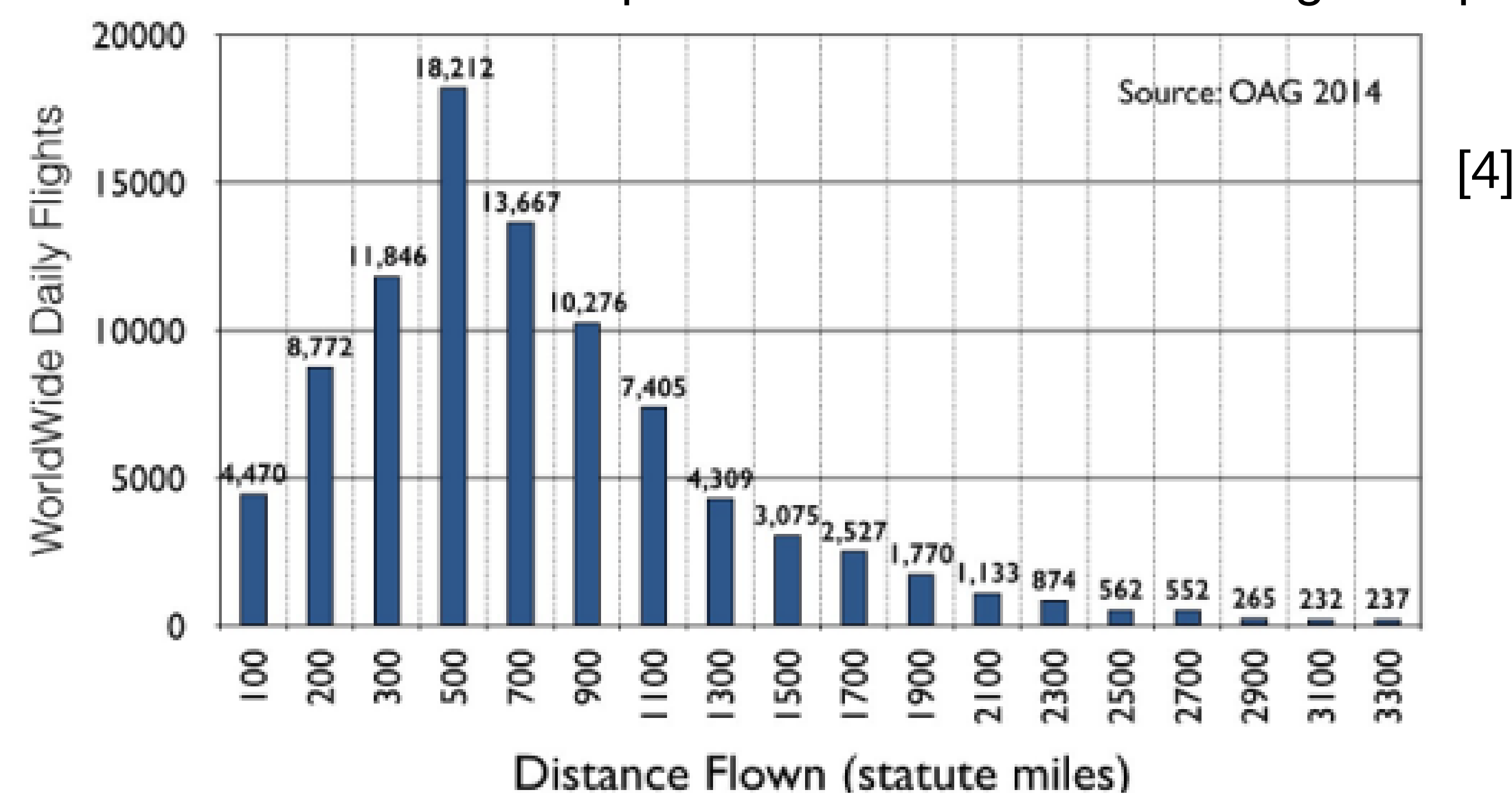
- It is crucial for the industry to recognize that women face unique challenges and barriers that may discourage them from pursuing careers in aviation.
- Lack of female representation can make many women hesitant to enter a male-dominated industry with historically limited growth opportunities for women.
- Some key challenges and barriers faced by women in aviation include:
  - i. Differential cost of training
  - ii. Existence of a glass ceiling
  - iii. Lack of role models and mentors
  - iv. Pervasive stereotyping and discrimination



## Green Aviation

### Key Facts

- 2.5% of global anthropogenic CO<sub>2</sub> emissions come from commercial aviation [3]
- Short and medium range flights under 3, 000 miles are the most prominent length
- Pilots need to become “rated” for specific aircrafts after receiving their pilot’s licenses



### Hydrogen

- Airbus' ZEROe program aims to have hydrogen aircrafts ready for 2035 [5]
  - Aim to reach 1000–2500 mile ranges.
  - 100-to-200 passenger capacity.
  - These smaller planes can be speculated to have cheaper flight training costs
- Hydrogen can be produced in a variety of different ways with varying carbon emissions, commonly referred to as the colours of hydrogen.
  - 95% of hydrogen in the US is still produced from steam methane reforming
  - Electrolysis is a method where electricity can be run through water to produce hydrogen gas for later usage in a fuel cell

## Conclusion and Future Steps

### Conclusions

- Prime opportunity to encourage young women to join growing aviation industry, to help alleviate labour concerns amidst retiring workforce and tackle high gender disparity.
- As the hydrogen industry continues to develop, green aviation is speculated to become more and more popular allowing for an increased number of shorter-ranged flights.
- Green aviation will provide many more opportunity for young women, however it is not expected to get rid of the systemic barriers within the industry, such as male-dominated culture, lack of role models, stereotyping and discrimination

### Future Steps

- Investigate enrollment trends for aviation-related programs at post-second institutions.
- Examine policy regarding short-haul and long-haul flights, look towards cases where short-haul flights may be limited due to environmental concerns.
- Determine interest levels of large airlines in adoption green aviation, as well as specific details on how green aviation may change flight (e.g., size of fleets and number of flights).
- Analyze gender disparity within levels of flying using flight certification data (the highest credential held by a pilot at an airline) – use findings to determine in which airlines depict the least disparity, discuss which behaviours may promote gender equity.

### References

- [1] 2022 Women in Aviation Advisory Board Report, [https://www.faa.gov/regulations\\_policies/rulemaking/committees/documents/media/WIAB\\_Recommendations\\_Report\\_March\\_2022.pdf](https://www.faa.gov/regulations_policies/rulemaking/committees/documents/media/WIAB_Recommendations_Report_March_2022.pdf)
- [2] Aag-workforce-demographics.pdf, <https://www.aa.com/content/images/customer-service/about-us/corporate-governance/aag-workforce-demographics.pdf>
- [3] D. S. Lee et al., “The contribution of global aviation to anthropogenic climate forcing for 2000 to 2018,” Atmos Environ, vol. 244, p. 117834, Jan. 2021, doi: 10.1016/j.atmosenv.2020.117834.
- [4] Marien, Ty & Antcliff, Kevin & Guynn, Mark & Wells, Douglas & Schneider, Steven & Tong, Mike & Trani, Antonio & Hinze, Nicholas & Dollyhigh, Samuel. (2018). Short-Haul Revitalization Study Final Report. 10.13140/RG.2.2.10821.37608.
- [5] “Zeroe,” Airbus, <https://www.airbus.com/en/innovation/low-carbon-aviation/hydrogen/zeroe>.

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